

UniCore Campus Operating Platform

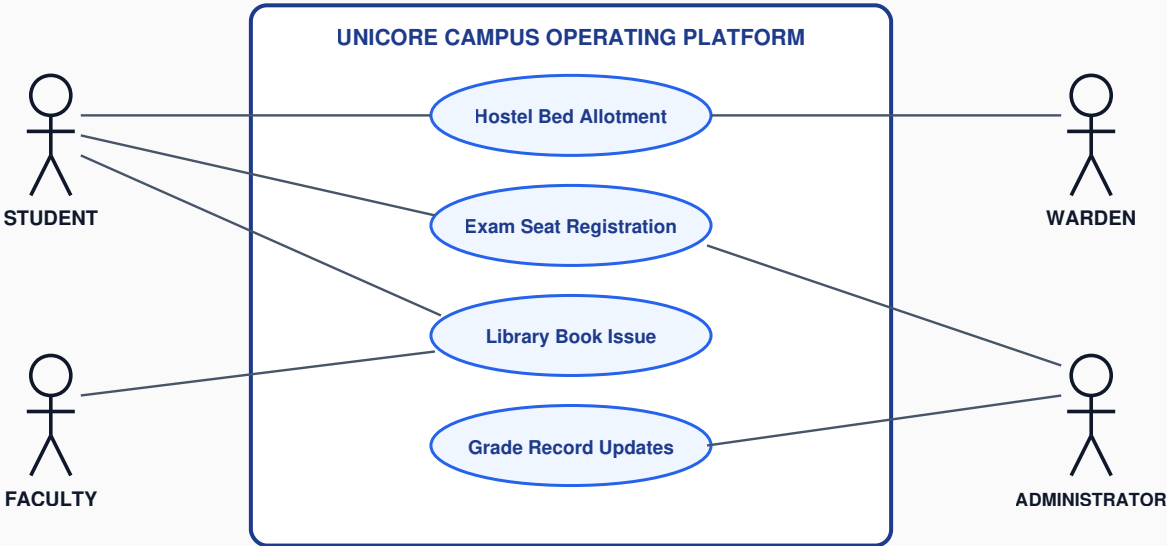
System Architecture Specification — Level 0 & Level 1 UML & DFD Documentation

Document Type: Technical Architecture PDF **Target System:** UniCore Campus Platform **Specification Levels:** Level 0 & Level 1 (UML / DFD)

1. UML LEVEL 0 — USE CASE DIAGRAM

Use Case Architecture Overview

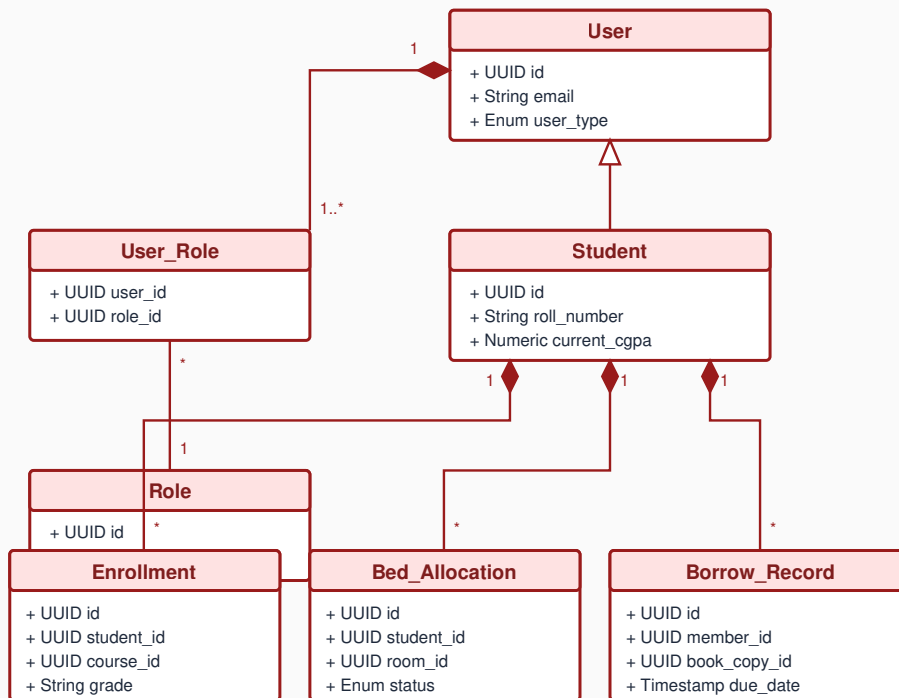
The Level 0 Use Case Diagram maps core primary actors (Student, Warden, Faculty, Administrator) to their designated operational interactions within the UniCore system boundary.



2. UML LEVEL 1 — CLASS DIAGRAM (DOMAIN MODEL)

Domain Class Architecture & Entities

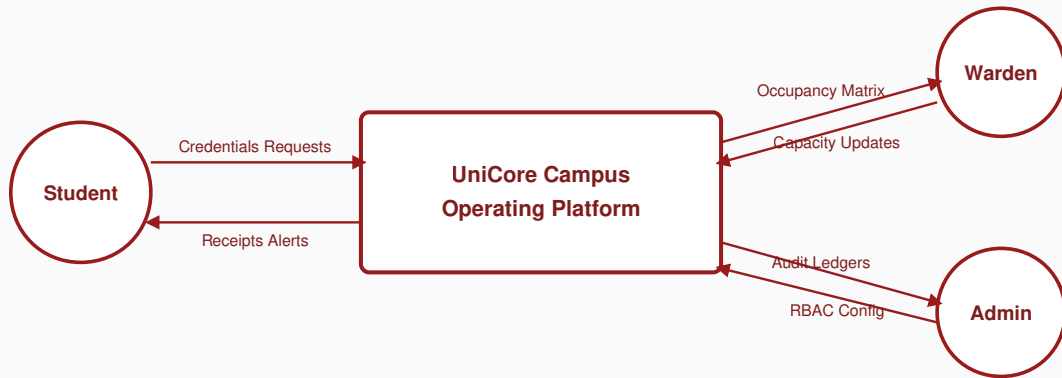
The Level 1 Class Diagram details core entity attributes, inheritance hierarchies (User → Student), compositions, and primary associations across modules.



3. DFD LEVEL 0 — CONTEXT DIAGRAM

System Context & External Entities Flow

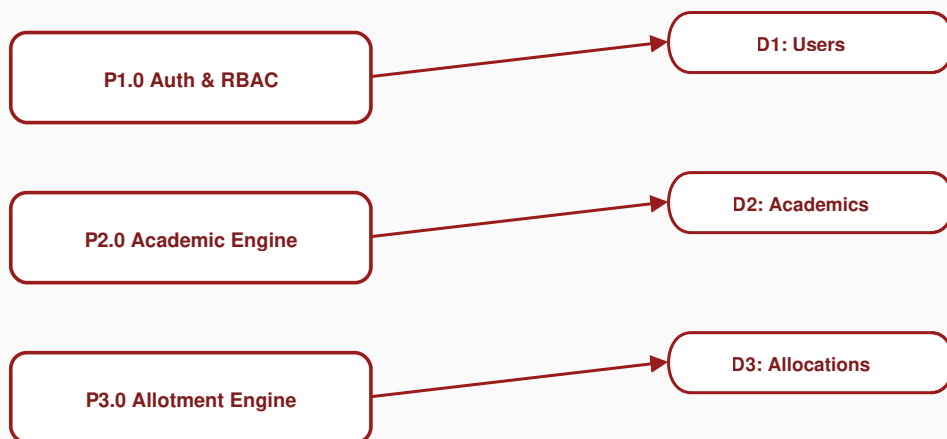
The Level 0 Data Flow Diagram illustrates high-level inbound and outbound operational data flows between external system entities and the UniCore core boundary.



4. DFD LEVEL 1 — PROCESS DECOMPOSITION

Subsystem Decomposition & Data Store Mapping

The Level 1 Data Flow Diagram decomposes core processing engines and maps their primary output states into corresponding system data stores.



5. ARCHITECTURAL DATA MAPPING REFERENCE

Level / Diagram	Core Components	Data Inputs / Interacting Entities	Target Data Store / Output
UML L0 Use Case	4 Primary Operational Use Cases	Student, Warden, Faculty, Administrator	Operational Modules Interaction
UML L1 Class Diagram	7 Key Classes (User, Student, Roles, etc.)	UUIDs, Enums, Strings, Timestamps	Relational Schema Representation
DFD L0 Context Diagram	Central UniCore System Boundary	Credentials, Capacity Updates, RBAC Config	Receipts, Occupancy Matrix, Audit Ledgers
DFD L1 Process Diagram	P1.0 Auth, P2.0 Academic, P3.0 Allotment	Decomposed Processing Logic	D1: Users, D2: Academics, D3: Allocations